

GDPS Metro Global – GM Workshop

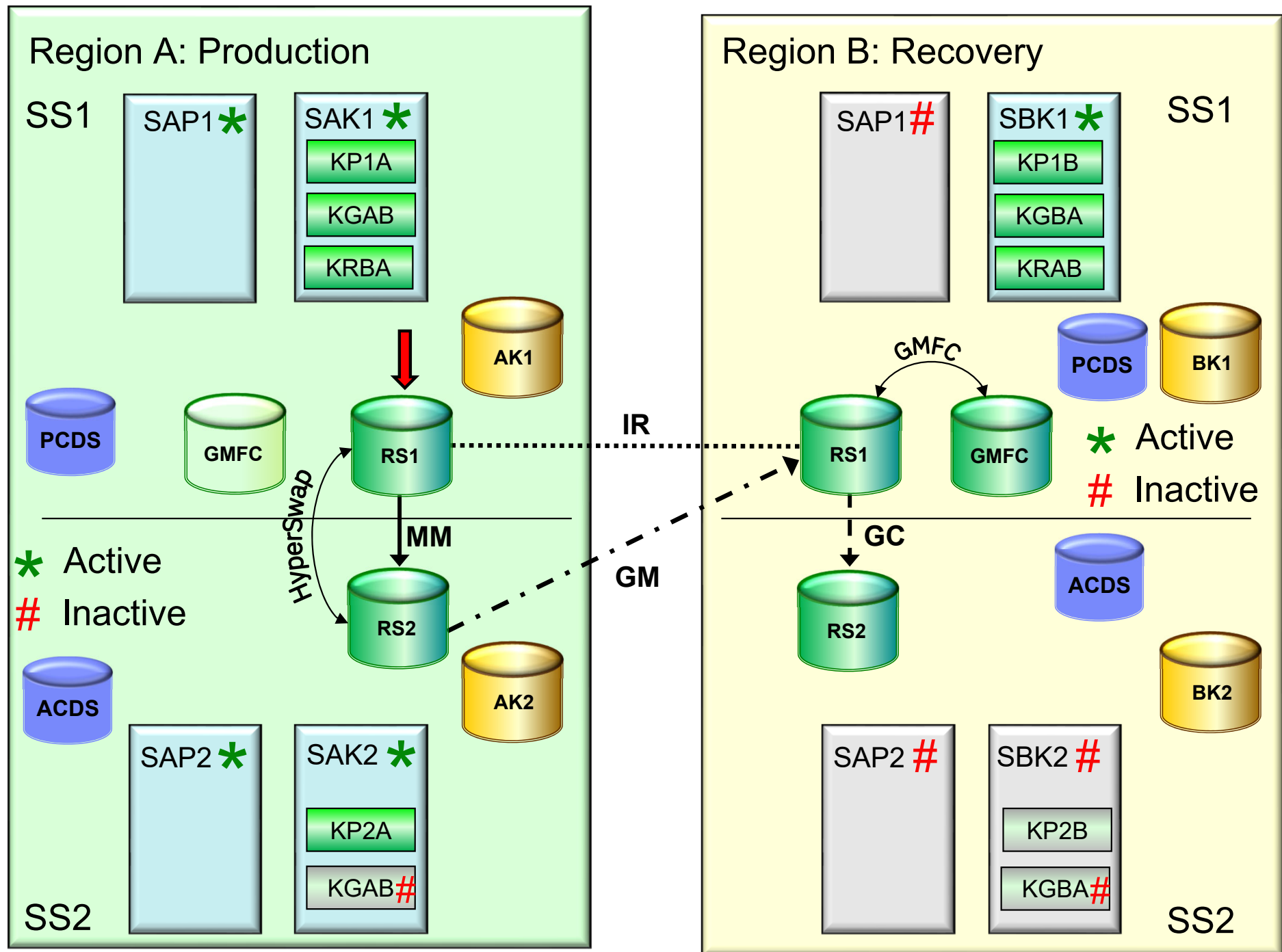
GDPS METRO GLOBAL – GM CONFIGURATION CONTROLLING SYSTEM AND COUPLE DATA SET SETUP



Agenda

- Controlling systems in an MGM
- Couple Data Set (CDS) setup

GDPS MGM 4-site K Sys setup (cascaded mode)



GDPS MGM Controlling Systems

GDPS Metro

- **Site2 Kp-sys** running on a Site2 server from local disk in Site2 Region A and Region B
 - Also referred to as **Kp2**
 - Only usually active in Application region
- **Site1 Kp-sys** running on a Site1 server from local disk in Site1 Region A and Region B
 - Also referred to as **Kp1**
 - Usually active in both regions

GDPS Global – GM

- **Kg-sys “function”** running on non-PPRCed disk, *embedded* in one of the Kp-sys
 - Under a second NetView instance, separate from the NetView running the Kp-sys GDPS Metro code)
 - Can embed in either of the Kp-sys - Recommend embedding in Kp-sys in opposite site to GM primary disk
 - Strong recommendation in making Kg-sys function moveable between Site1 and Site2 Kp-sys in both regions

GDPS MGM Controlling Systems

GDPS Global – GM (cont'd)

- Kr-sys in R-site can be
 - Also referred to as Kr or R-sys
 - Can be stand-alone system in R-site or co-located with another GDPS/GM NetView instance for R-sys (can run in XCFLOCAL mode for simplicity)
 - Other than for outages, Kr-sys in both regions must be active at all times

GDPS MGM Controlling Systems summary (cascaded mode)

Region A	GDPS Function	Region
SS1 Controlling system running	Kp-sys function for region ... Kg-sys function for region ... (recommended moveable to SS2) Kr-sys function for region ...	A A B
SS2 Controlling system running	Kp-sys Kg-sys (moveable here)	A A
Region B	GDPS Function	Region
SS1 Controlling system running	Kp-sys function for region ... Kg-sys function for region ... (recommended moveable to SS2) Kr-sys function for region ...	B B A
SS2 Controlling system running	Kp-sys Kg-sys (moveable here)	B B

- The Region A Kp-sys/Kg-sys/Kr-sys functions manage PPRC within Region A and manage GM from Region A to Region B and vice versa for the Region B functions

Production versus recovery mode for Controlling systems

- Can run in either production or recovery mode
- Requires two different CDS sets in each region
 - When in production mode it is npart of the production Sysplex using the production CDSs
 - When in recovery mode, it is not part of the production Sysplex and uses a separate set of CDSs (recovery CDS set)
- Switching between modes requires different set of load parameters
 - Different system names could be used depending on mode to aid identification

Kp/Kg-sys considerations

- Controlling system that houses the combined Kp/Kg system (normally Kp1/Kg) should be set up to run using ALLOWACCESS(NO) in IXGCNFxx to ensure no allocation against the LOGR CDS or use the LOGRY/Z support in GDPS 4.2 and higher and z/OS 2.2 and higher if K-sys requires use of LOGR for client reasons.

Moving the Kg-Sys

- The Kg-Sys instance should run in:
 - Site1 Controlling system in an MGM cascaded configuration
 - Site2 Controlling system in an MGM multi-target configuration
- Several scenarios require to move the Kg-Sys instance to the other site to remain capable of reconfiguring MGM
- These steps outline what is necessary to move the Kg-Sys instance:
 - Prepare in advance alternate GEOGROUP member on every K-Sys (rename old)
 - Contains alternate TCPIP communication details for Kg-Sys to Kr-Sys
 - Contains alternate Serversite details for the alternate Kp-Sys

Moving the Kg-Sys (cont'd)

With APAR PH54754 installed

- Copy alternate GEOGROUP into place in all systems, retaining the old version
- Shutdown Kg-Netview
- Issue in Netview command GDPSSITE REFRESH in the Master Kp-Sys and validate the SITETABLE entries
- Refresh the GEOGROUP in Master Kp-Sys to update the Kg comms definitions
- Issue in Netview command GDPSSITE REFRESH in the Alternate Kp-Sys and validate the SITETABLE entries
- Refresh the GEOGROUP in Master Kp-Sys to update the Kg comms definitions
- Start Kg-Sys NetView in alternate location, validate both the GEOGROUP Sitetable and comms status
- Validate new SITETABLE definitions are now also in place in the Kr-Netview
- Refresh the GEOGROUP in Kr-Netview to update the comms

Couple Data Set (CDS) setup for GDPS MGM

- GDPS MGM *must* use pre-allocated CDS in each region
- Replication of CDS from both Metro sites is not possible without risk:
 - If replication is from both sites, failure in one site would prevent consistency group (CG) formation for all disk in both sites
 - If only one CDS is replicated (asymmetric support) then the risk is that the CDS I/O was incomplete during CG formation resulting in a broken CDS that could not be used for restart (only one CDS of each type is replicated)
- Implications:
 - Additional systems management tasks to ensure updates to CDS (including content eg: CFRM and WLM policy changes) in one region are reflected in other region
 - Special considerations for both MQ and RLS exploiters as they have stored information in the XCF CDS
- Can use same Production CDS names in both regions
 - Same COUPLExx CDS definitions used in Region A production can be used for
 - Region B production (IPL on RegB.RS1)
 - Region B testing (IPL on FC1/X-disk)

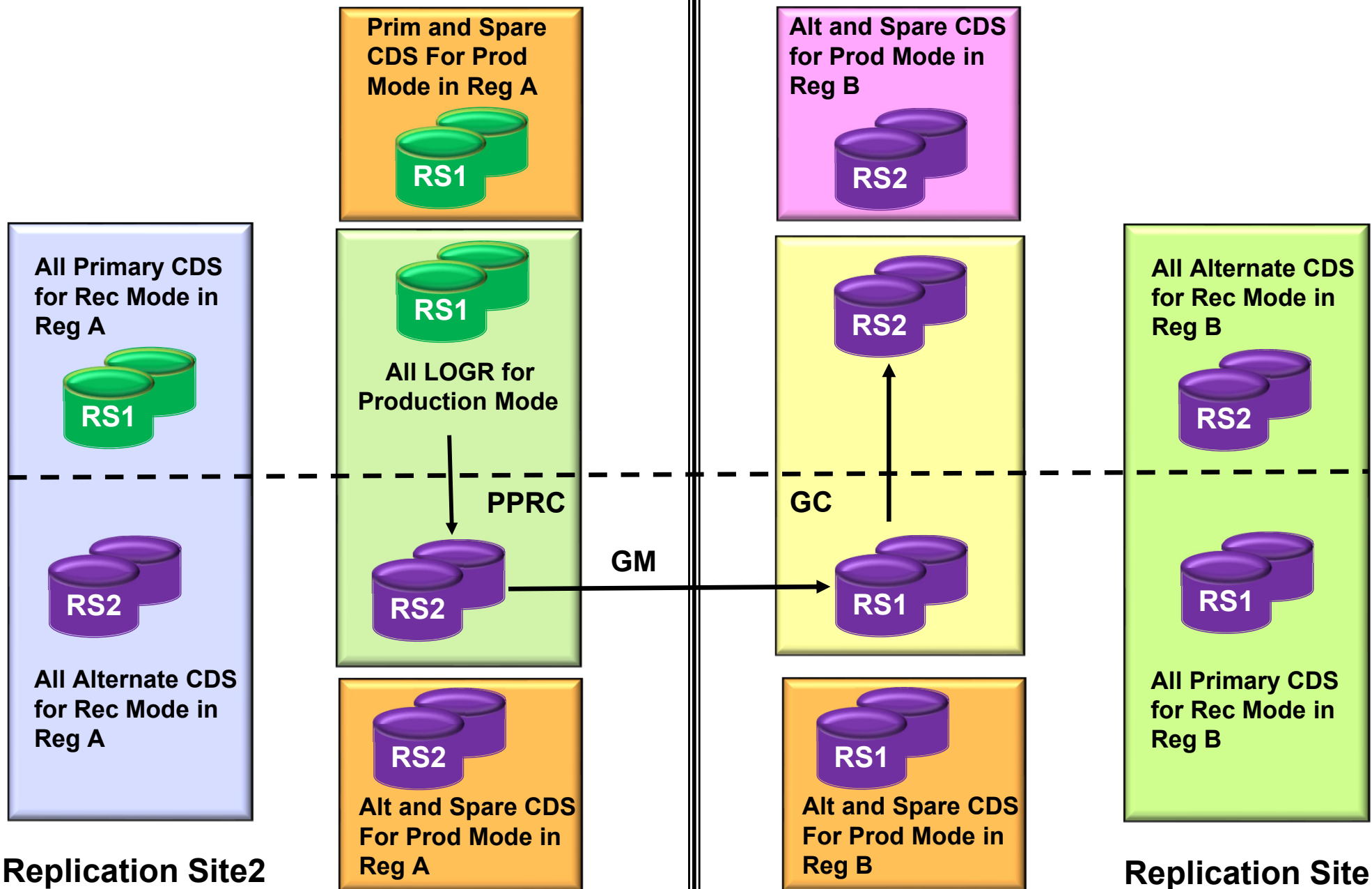
CDS setup

Replication Site1

Region A

Region B

Replication Site2



END OF LECTURE